

DILLAN IMANS

Medical AI Researcher

Relocating to Munich, Germany (Oct 2026)
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SUMMARY

Medical AI Researcher with first-author publications at IEEE BIBM 2025 and Diagnostics, specializing in multimodal learning and clinical AI deployment across diagnostic imaging and surgical vision, with additional experience in genomics and proteomics. Incoming M.Sc. AI in Biomedicine at the Technical University of Munich.

EDUCATION

M.Sc. AI in Biomedicine, Technical University of Munich	Oct 2026 - Oct 2028 (Expected)
B.S. Computer Science & Engineering, Sungkyunkwan University	Aug 2022 - Aug 2026
95.7 / 100 · Dean's List 2023 · Academic Excellence Scholarship	

EXPERIENCE

AI Research Engineer · Superintelligence Lab SKKU, South Korea	Oct 2024 - Present
- Developed unsupervised domain adaptation framework for brain tumor segmentation across MRI domain shift, outperforming benchmarks by ~5–10% AUC (First Author, IEEE BIBM 2025) - Built end-to-end retinal analysis system covering optic disc/cup segmentation, vessel analysis, and glaucoma detection; optimized for clinical deployment via ONNX with low-latency inference, integrated with industry-partnered screening platform. - Designed cross-modality knowledge distillation framework for hypertension risk prediction from unpaired MRI, ECG, and fundus images (First Author, preprint available)	
AI Research Engineer · Labren CUHK, Hong Kong	Jun 2024 - Present
- Designed preprocessing infrastructure and annotation QA pipeline for the first large-scale surgical action recognition dataset (11,915 clips) - Collaborated with NVIDIA on multimodal surgical robotics dataset construction, supporting pipeline architecture and data standardization - Co-authored 3 manuscripts in submission at npj Digital Surgery and top-tier conferences	
AI Research Engineer · Infolab SKKU, South Korea	Feb 2024 - Oct 2024
- Developed explainable dynamic ensemble framework for depression detection and severity prediction, achieving 88.3% accuracy across 2,763 clinical records with SHAP interpretability (First Author, Diagnostics 2024) - Conducted statistical analysis and model benchmarking across 20+ classical, ensemble, and dynamic selection algorithms for mental health screening	

PUBLICATIONS

Full list: scholar.google.com/citations?user=o-TtUq0AAAAJ

- Imans D. et al. "Unsupervised Domain Adaptation with SAM-RefiSeR for Enhanced Brain Tumor Segmentation." IEEE BIBM 2025. (First Author)
- Imans D. et al. "Explainable Multi-Layer Dynamic Ensemble Framework Optimized for Depression Detection and Severity Assessment." Diagnostics, 14(21), 2385 (2024). (First Author)
- 1 first-author preprint on arXiv · 3 co-authored preprints under review

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, JavaScript
ML / AI: PyTorch, MONAI, HuggingFace, scikit-learn, OpenCV, wandb, TensorBoard
Medical / Bio: NiBabel, SimpleITK, pydicom, AlphaFold, BLAST
Tools: Git, Linux, Docker, AWS
Spoken Languages: English & Indonesian (Fluent), Mandarin & Korean (Conversational)